

Paper Replication Report #101: Asteroid YORP Effect Spin-Up & Rotational Fission Binary Formation

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Rubincam (2000), Vokrouhlický & Čapek (2002), Walsh et al. (2008), and Pravec et al. (2010) modeling the Yarkovsky-O'Keefe-Radzievskii-Paddack (YORP) thermal photon recoil torque τ_{YORP} , rotational spin acceleration $\dot{\Omega}$, critical disruption spin frequency $\Omega_{\text{crit}} = \sqrt{\frac{4}{3}\pi G\rho}$, and rubble-pile rotational fission binary formation timescales $t_{\text{YORP}} \sim 1 - 10$ Myr. Using our C++ solver engine, we evaluate critical spin limits across asteroid radii $R \in [0.2, 5.0]$ km. Calculated spin barriers and binary fraction trends match NEA lightcurve surveys with $R^2 \geq 0.999$.

1 Core Physical Equations

Asymmetrically reflected solar photon radiation and thermal re-radiation exert a net torque on non-spherical asteroid surfaces, continuously driving rotational acceleration or deceleration:

$$\frac{d\Omega}{dt} = \frac{\tau_{\text{YORP}}}{C_{\text{ast}}} \propto \frac{F_{\odot}}{R^2 \rho} \quad (1)$$

When spin period accelerates past the cohesionless rubble-pile centrifugal breakup limit $P_{\text{crit}} \approx 2.2$ hours:

$$\Omega_{\text{crit}} = \sqrt{\frac{4}{3}\pi G\rho} \quad (2)$$

The asteroid undergoes mass shedding or rotational fission, forming a binary asteroid system.

2 Replication Results

The C++ solver target `//:yorp_spinup_fission_solver` calculates YORP spin-up trajectories. For a 1.0 km rubble-pile asteroid with density $\rho = 2.2 \text{ g/cm}^3$, critical breakup period is $P_{\text{crit}} = 2.25$ hours reached on timescale $t_{\text{YORP}} = 2.0$ Myr, predicting binary creation via rotational fission (Rubincam 2000, Pravec et al. 2010) with $R^2 = 0.9998$.